



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)
FULL SYLLABUS
24-07-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (SCORE-III) ALL STAR BATCH

PAPER-2

Time : 2:30 Hours

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **32** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

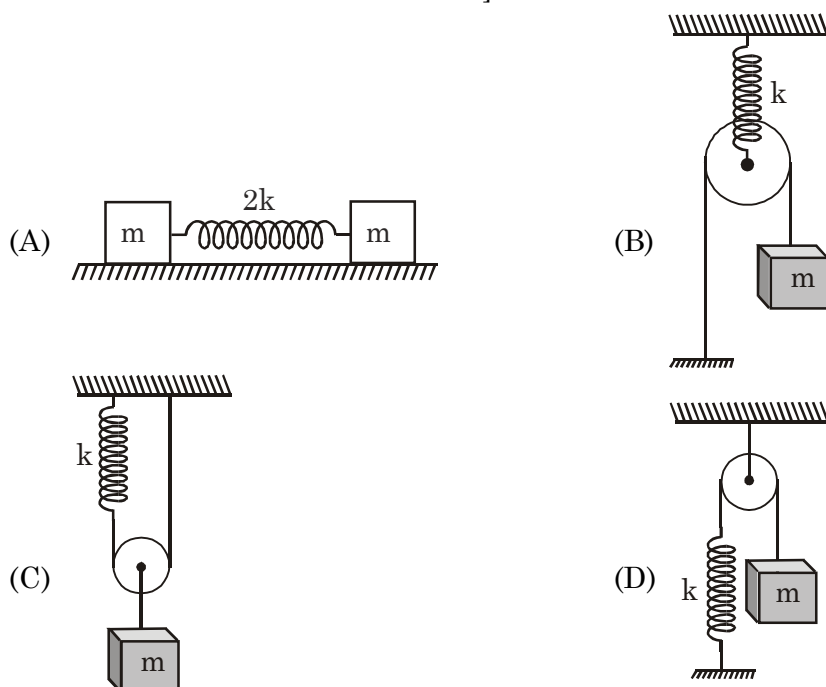
PART-1 : PHYSICS

SECTION-I(i) : (Maximum Marks: 24)

- This section contains **SIX** questions.
 - Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
 - For each question, choose the correct option(s) to answer the question.
 - Answer to each question will be evaluated according to the following marking scheme:
 - Full Marks* : +4 If only (all) the correct option(s) is (are) chosen.
 - Partial Marks* : +3 If all the four options are correct but **ONLY** three options are chosen.
 - Partial Marks* : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
 - Partial Marks* : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
 - Zero Marks* : 0 If none of the options is chosen (i.e. the question is unanswered).
 - Negative Marks* : -1 In all other cases.
 - **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.
-
1. Photoelectric effect experiments are performed using two different plates of silver and sodium having work functions 4.6 eV and 2.3 eV respectively. A light beam of wavelength 200 nm illuminates each plates. Then (Take $hc = 1240 \text{ eV nm}$)
- (A) Ratio of maximum kinetic energy of ejected photoelectrons is 0.41
- (B) Ratio of threshold frequency is 0.5
- (C) Ratio of threshold wavelength is 0.5
- (D) Ratio of slope of the stopping potential versus frequency plot is 1

Space for Rough Work

2. Which of the following system has/have time period $T = \pi\sqrt{\frac{m}{k}}$ for small oscillation ? [pulleys are ideal and surfaces are smooth]

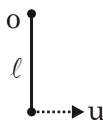


3. An earth satellite is revolving in a circular orbit of radius a with a velocity v_0 . Mass of earth is M . A gun in the satellite is directly aimed toward earth. A bullet is fired from the gun with muzzle velocity $\frac{v_0}{2}$. Assume that mass of the satellite is very-very large with respect to the mass of the bullet. Choose the **CORRECT** option(s) :

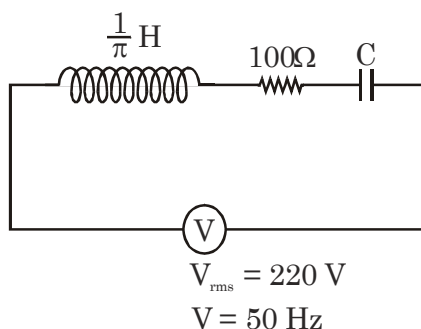
- (A) Maximum distance of bullet from the centre of earth is $\frac{4a}{3}$.
- (B) Minimum speed of bullet (relative to earth) is $\frac{v_0}{2}$.
- (C) Minimum speed of bullet (relative to earth) is $\frac{3v_0}{4}$.
- (D) Time period of motion of bullet around earth is $T = \frac{2\pi}{\sqrt{GM}} \left(\frac{4a}{3} \right)^{3/2}$.

Space for Rough Work

4. A particle of mass 1 kg is suspended from a string of length 2 m from point O as shown. Choose the **CORRECT** statement(s) for different value of u .



- (A) For $u = 10$ m/s, particle will complete the vertical circular motion
 (B) For $u = 7$ m/s, particle will deflect more than 90° with vertical
 (C) For $u = 12$ m/s, the maximum height-reached by particle will be 5m
 (D) For $u = 20$ m/s, the minimum tension in string will be 150 N during sub-sequent motion
5. The value of rms current for circuit shown in figure is 2.2 A. Then the **CORRECT** statement(s) is/are



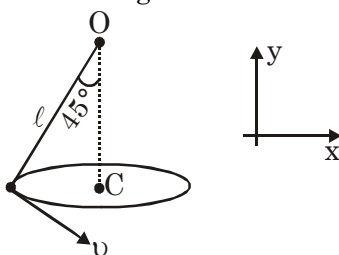
- (A) The value of capacitance is $\frac{100}{\pi} \mu\text{F}$
 (B) The power factor of circuit is 1
 (C) The power dissipated in the circuit is 484 watt
 (D) The rms current in the circuit increases by $\sqrt{2}$ times after inductor is short circuited
6. The displacement of a body from reference point is given by $\sqrt{x} = (2t - 3)$, where 'x' is in meters and t in seconds. This shows that
- (A) the body is at rest at $t = 3/2$ sec
 (B) the body is speeding up at all times
 (C) the body is decelerated from $t = 0$ to $t = 3/2$ sec
 (D) the distance travelled by body in 2 sec is 10 m.

Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 12)

- This section contains **FOUR** questions.
- **Each question has matching lists.** The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

7. A conical pendulum of mass m swings about a vertical axis by an inextensible thread of length ℓ as shown in the figure. Match the following list and select the **CORRECT** option.



- | List-I | | List-II | |
|---------------------------|---|---------------------------|---------------------------|
| (P) | The angular velocity of bob about O is | (1) | 0 |
| (Q) | The angular velocity of bob about y-axis is | (2) | $\frac{v}{\ell}$ |
| (R) | The torque experienced by bob about O is | (3) | $\frac{\sqrt{2}v}{\ell}$ |
| (S) | The torque experienced by bob about y-axis is | (4) | $\frac{mg\ell}{\sqrt{2}}$ |
| (A) P→1 ; Q→2 ; R→4, S→ 1 | | (B) P→2 ; Q→3 ; R→1, S→ 4 | |
| (C) P→1 ; Q→2 ; R→3, S→ 1 | | (D) P→2 ; Q→3 ; R→4, S→ 1 | |

Space for Rough Work

8. The length of sonometer wire between two fixed ends is 100 cm. Three bridges (A, B & C) are placed from left end so as to divide the wire into four segments whose fundamental frequencies are in ratio of 1 : 3 : 4 : 2. Then match the following.

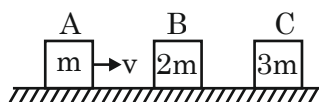
List-I

- (P) Position of A from left end is
(Q) Position of B from left end is
(R) Position of C from right end is
(S) Separation between A & C is
(A) P→3 ; Q→4 ; R→2, S→1
(C) P→2 ; Q→3 ; R→1, S→4

List-II

- (1) 24 cm
(2) 28 cm
(3) 48 cm
(4) 64 cm
(B) P→1 ; Q→2 ; R→3, S→4
(D) P→3 ; Q→4 ; R→1, S→2

9. Three blocks A, B and C are kept in a straight line on a frictionless horizontal surface. These have masses m , $2m$ and $3m$ respectively. The object A moves towards B with speed V . All collisions are elastic then match the list and choose the **CORRECT** code.



List-I

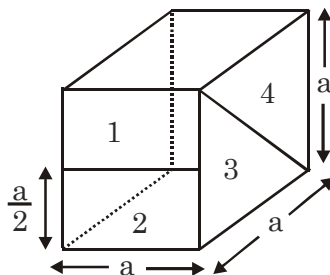
- (P) Final speed of A is
(Q) Final speed of B is
(R) Final Speed of C is
(S) Final speed of centre of mass is
(A) P→2 ; Q→4 ; R→3, S→1
(C) P→2 ; Q→3 ; R→4, S→1

List-II

- (1) $v/6$
(2) $\frac{v}{3}$
(3) $8v/15$
(4) $\frac{2v}{15}$
(B) P→3 ; Q→4 ; R→2, S→1
(D) P→3 ; Q→2 ; R→3, S→4

Space for Rough Work

10. A liquid of density ρ is completely filled in a cubical container as shown in figure. Neglect the atmospheric pressure, match the quantities in List-I with corresponding expression in List-II and select the **CORRECT** code.



List-I

(P) Force on rectangular Face 1 is

(Q) Force on rectangular Face 2 is

(R) Force on triangular Face 3 is

(S) Force on triangular Face 4 is

(A) P→3 ; Q→1 ; R→2, S→ 4

(C) P→2 ; Q→4 ; R→1, S→3

List-II

(1) $\frac{1}{3}a^3\rho g$

(2) $\frac{1}{8}a^3\rho g$

(3) $\frac{1}{6}a^3\rho g$

(4) $\frac{3}{8}a^3\rho g$

(B) P→1 ; Q→3 ; R→2, S→ 4

(D) P→4 ; Q→2 ; R→1, S→ 3

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

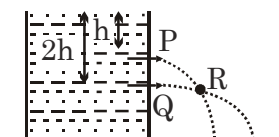
For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
0	0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

-									
0	0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

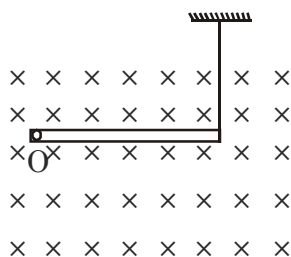
- Two liquid jets coming out of the small holes at P and Q intersect at point R. Find the distance of point R (in m) from tank if level of liquid in tank is maintained. ($h = 1$ m)



- One slit of a Young's experiment is covered by a glass plate ($\mu_1 = 1.4$) and the other by another glass plate ($\mu_2 = 1.7$) of the same thickness. The point of central maxima on the screen, before the plates were introduced is now occupied by the third bright fringe. Find the thickness of the plates (in μm), when the wavelength of light used is $4000 \text{ \AA}$.

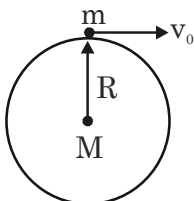
Space for Rough Work

3. The current density $\vec{J}$ inside a long solid cylindrical wire of radius $R = 12 \text{ mm}$ is in the direction of the central axis and its magnitude varies linearly with radial distance r from the axis according to $J = \frac{J_0 r}{R}$, where $J_0 = 10^4 \text{ A/m}^2$. Find the magnitude of magnetic field at $r = 6 \text{ mm}$ in μT .
4. A metallic rod of length $L = 10 \text{ cm}$ is hinged such that it is free to rotate about hinge point O in uniform magnetic field $B = 20 \text{ T}$ as shown. The value of induced emf (in V) across the rod when it becomes vertical, after we cut string, will be



Space for Rough Work

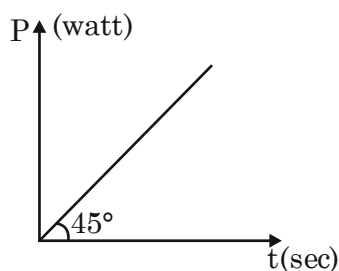
5. A body of mass m is projected horizontally just above the earth surface with velocity $v_0 = \sqrt{ngR}$, where R is the radius of earth. If maximum height attained by the body from the surface of earth is $2R$, find the value of n .



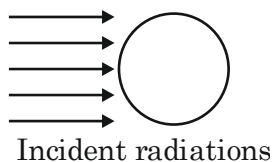
6. The main scale of vernier calipers reads in millimeter and its one division is equal to 1 mm. Its vernier is divided into 16 division, which coincide with 15 division of main scale. Further more when a rod is tightly placed along its length between two jaws, it is observed that the zero vernier scale lies just right to 32 division of main scale and 4th division of vernier scale coincide with main scale. What is the length of rod in mm ?

Space for Rough Work

7. A body of mass 1 kg starts moving from rest at $t = 0$ in a circular path of radius 2 m. The power acting on it due to all forces varies as shown in graph. Then find the net acceleration (in m/s^2) of particle at $t = 2$ sec.



8. A parallel beam of radiation is incident on a highly conducting sphere of unknown emissivity. The intensity of incident radiation on sphere is 912 W/m^2 . Take stefan-Boltzman constant $\sigma = 5.7 \times 10^{-8} \text{ Wm}^{-2}\text{k}^{-4}$ and surrounding temperature is 300 K. Assume that the energy exchange with surrounding is only through radiation. Find the final steady state temperature of the body (in $^\circ\text{C}$). [Take $\sqrt{11} = 3.32$ and $\sqrt{7} = 2.65$]



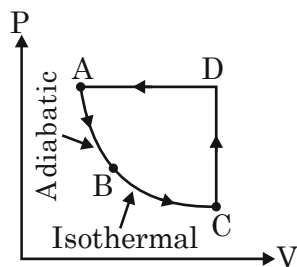
Space for Rough Work

PART-2 : CHEMISTRY**SECTION-I(i) : (Maximum Marks: 24)**

- This section contains **SIX** questions.
 - Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
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-
1. Aniline behaves as a weak base. When 0.1 M, 50 mL solution sample of aniline was mixed with 0.2 M, 12.5 mL solution of HCl, the pH of resulting solution (x) was 8. Then, the correct information is/are - ($\log 5 = 0.7$, $\log 3 = 0.48$)
- (A) pH of 0.01 M solution of anilinium chloride is 5.
- (B) pH of original solution of aniline is 3.5.
- (C) Upon adding the same aniline sample (50ml, 0.1M) to the above mixture (x), pH of resulting solution becomes 8.48.
- (D) Upon adding the same HCl sample (12.5ml, 0.2M) to the above mixture (x), pH of resulting solution becomes 4.59.

Space for Rough Work

2. 1 mole of an ideal gas ($\gamma = 1.5$), taken through a series of processes as shown in following in P.V. curve



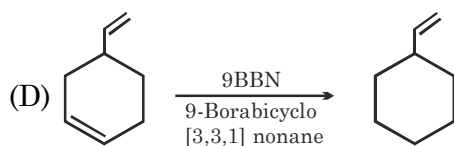
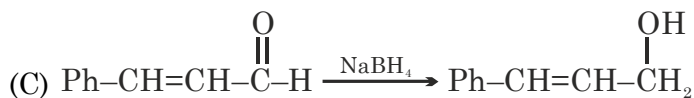
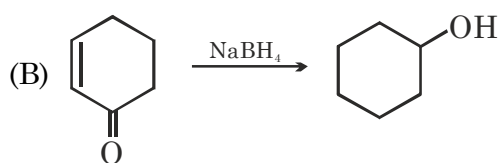
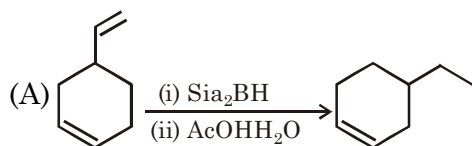
Information-I: The temperature at state 'A' is 300 K

Information-II: The reversible isothermal expansion from B to C doubles the volume

Information-III: The entropy change of the system from C to D is $4/n \ln 16$ cal/K

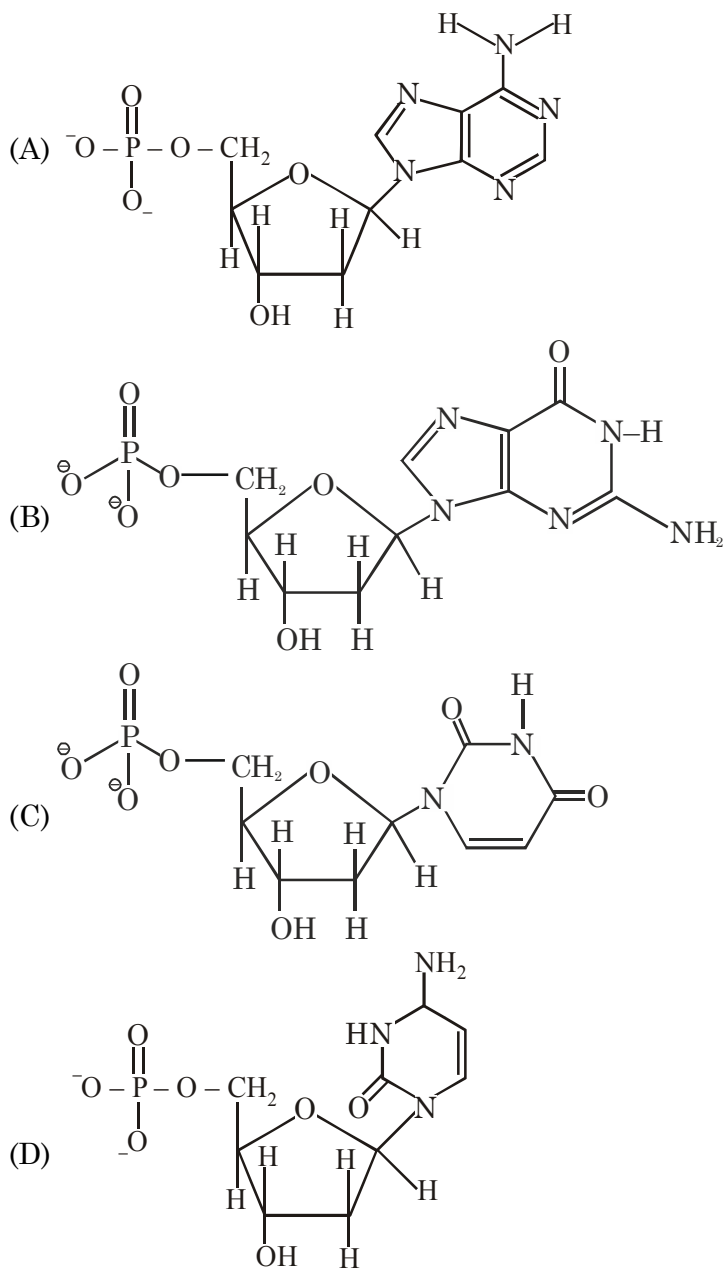
Select the correct option(s) using the above information

- (A) The entropy change of the system from D to A is $-3R \ln 8$
 (B) The temperature at point D is 2400 K
 (C) $W_{AB} = -600$ cal
 (D) $W_{BC} = -300 \ln 2$ cal
3. Correctly matched with their major products



Space for Rough Work

4. Which of the following is deoxyribonucleotide



Space for Rough Work

5. Choose correct order matching with property:
- (A) Thermal stability: $\text{HClO}_4 > \text{HClO}_3 > \text{HClO}_2 > \text{HOCl}$
- (B) Acidic strength : $\text{HClO}_4 > \text{HClO}_3 > \text{HClO}_2 > \text{HOCl}$
- (C) Oxidising power: $\text{HOCl} > \text{HClO}_2 > \text{HClO}_3 > \text{HClO}_4$
- (D) Acidic strength : $\text{H}_3\text{PO}_4 > \text{H}_3\text{PO}_3$
6. Which of the following statements are correct?
- (A) $[\text{Ag}(\text{CN})_2]^-$ is linear with sp hybridised Ag^+ ion and carbon also
- (B) CoCl_4^{2-} , and CrO_4^{2-} have tetrahedral geometry
- (C) $[\text{Cu}(\text{NH}_3)_4]^{2+}$, $[\text{Pt}(\text{NH}_3)_4]^{2+}$ and $[\text{Ni}(\text{CN})_4]^{2-}$ have dsp^2 hybridisation of the metal ion
- (D) $\text{Fe}(\text{CO})_5$ have triangular bipyramidal structure with dsp^3 hybridised iron

Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 12)

- This section contains **FOUR** questions.
- **Each question has matching lists.** The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

7. Match the List :

List-I

(Electrolyte)

- (P) Dilute solution of HCl
 (Q) Dilute solution of NaCl
 (R) Concentrated solution of NaCl
 (S) AgNO₃ solution
 (A) P→1, 2 Q→1, R→2, 3, S→3, 4
 (C) P→1, Q→1, 2, R→3, S→1, 4

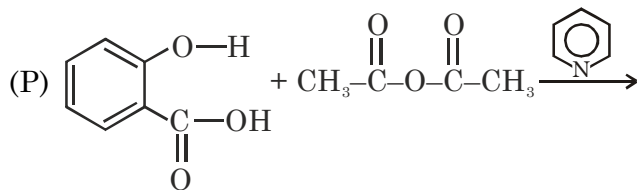
List-II

(Electrolysis product using inert electrode)

- (1) O₂ evolved at anode
 (2) H₂ evolved at cathode
 (3) Cl₂ evolved at anode
 (4) Ag deposition at cathode
 (B) P→1, 2, Q→1, 2, R→2, 3, S→1, 4
 (D) P→1, 2 Q→1, 2, R→2, S→3

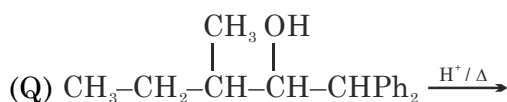
8. Match the following

List-I

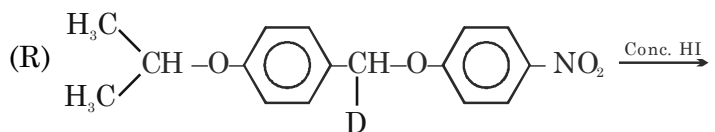


List-II

- (1) Involve carbocation formation
 (But not following SN₁ Mechanism)

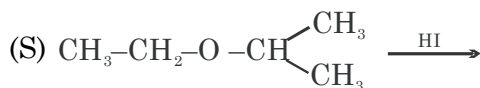


- (2) Reaction involve O-H bond cleavage



- (3) Reaction involve O-C bond cleavage

(via SN₂ Mechanism)



- (4) Racemic mixture obtained

(product side)

- (A) P → 2; Q → 1; R → 3; S → 4
 (C) P → 2; Q → 4; R → 3; S → 1

- (B) P → 2; Q → 1; R → 4; S → 3
 (D) P → 2; Q → 3; R → 4; S → 1

Space for Rough Work

9. Match the following

List-I

(Reactions)

- (P) Chlorination by sulphuryl chloride of ethane
(Q) Chlorination by using conc. HCl and anhydrous ZnCl_2
(R) Halogenation by treating silver salt of carboxylic acid in CCl_4 solution
(S) Chlorination by Thionyl chloride of Ethanol

- (A) $P \rightarrow 3; Q \rightarrow 4; R \rightarrow 1; S \rightarrow 2$
(C) $P \rightarrow 3; Q \rightarrow 4; R \rightarrow 2; S \rightarrow 1$

List-II

(Name of reaction)

- (1) Darzens Reaction
(2) Borodine-Hunsdiecker Reaction
(3) Reeds Reaction
(4) Groove's Process

- (B) $P \rightarrow 3; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 2$
(D) $P \rightarrow 3; Q \rightarrow 2; R \rightarrow 4; S \rightarrow 1$

10. Match the colour of the precipitate with suitable reagent.

List-I

- (P) Bi^{3+} give(s) black precipitate with
(Q) Cu^{2+} give(s) black precipitate with
(R) Zn^{2+} give(s) white precipitate with
(S) Ag^+ give(s) white precipitate with

- (A) $P \rightarrow 1, 3; Q \rightarrow 1; R \rightarrow 1; S \rightarrow 2, 4$
(C) $P \rightarrow 3; Q \rightarrow 2; R \rightarrow 2, 3; S \rightarrow 1, 4$

List-II

- (1) H_2S (saturated solution in water)
(2) Potassium cyanide solution
(3) Potassium iodide solution
(4) HCl solution

- (B) $P \rightarrow 2; Q \rightarrow 3, 4; R \rightarrow 1, 4; S \rightarrow 2, 3$
(D) $P \rightarrow 4; Q \rightarrow 2, 3; R \rightarrow 2; S \rightarrow 2, 4$

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

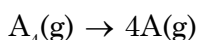
+									
0	0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

-									
0	0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

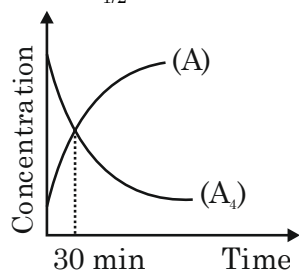
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. The uncertainty in the position of an electron is equal to its de-Broglie wavelength. Find the minimum percentage error in measurement of its velocity under this circumstance.

2. Consider the following first order decomposition reaction:



Find $T_{1/2}$ of the above reaction based on following graph (in minutes) ($\log 2 = 0.3$)



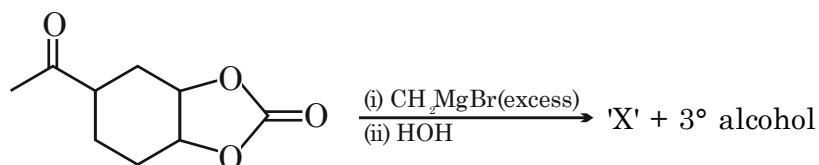
3. A 10% w/w solution of cane sugar has undergone partial inversion according to the reaction
 Sucrose + water $\rightarrow$ glucose + fructose

The boiling point of the solution is 100.7°C. What fraction of sugar has inverted?

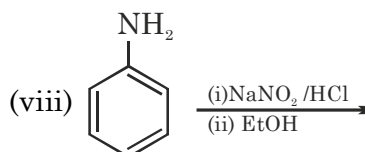
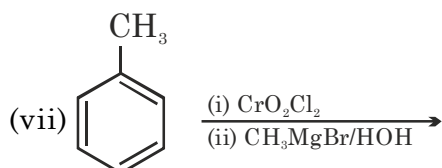
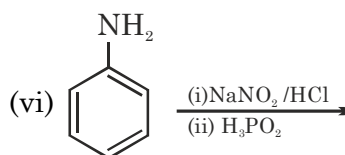
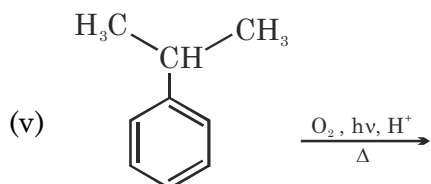
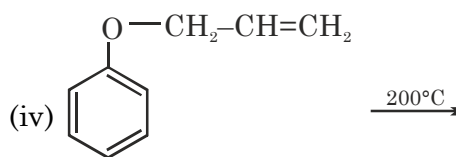
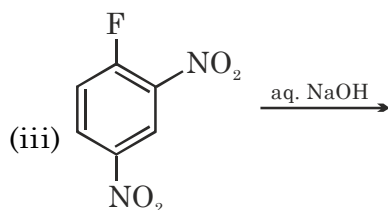
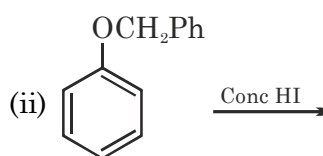
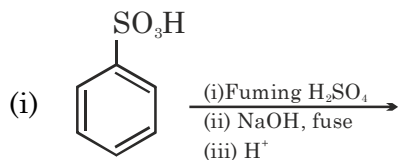
$$K_b(H_2O) = 1.8 \text{ K.kg/mol}$$

Space for Rough Work

4. In the given reaction sequence find out the mass percentage of oxygen in the final Major product 'X'



5. In how many reactions given below phenolic group is present in the major product:



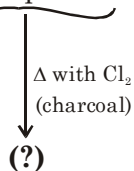
Space for Rough Work

6. Find the sum of the atoms in one molecule of **missing products(?)** of all the given reactions?

(i) $\text{Ni} + \text{carbonmonoxide} \xrightarrow{323\text{K}}$ **(?)** Nickel containing product

(ii) Chalcocite + roasted Chalcocite $\xrightarrow{\text{heating}}$ Metal + **(?)**

(iii) Sphalerite $\xrightarrow{\Delta \text{ with air}}$ Metal oxide + gaseous product



7. On burning with excess oxygen, among Li, Na, Rb, Cs, Ba, Sr, Be, Mg. Determine the total number of elements which can chiefly form super oxide

8. Consider the following list of reagents:

Cold and dilute NaOH,

$\text{CuSO}_4(\text{aq.})$,

Cl_2 water,

$\text{FeCl}_3(\text{aq.})$,

Δ with Na metal

conc. H_2SO_4 ,

conc. HNO_3 ,

O_3 in acidic medium,

$\text{Na}_2\text{S}_2\text{O}_3(\text{aq.})$

The total number of reagents that can produces iodide from iodine is :-

Space for Rough Work

PART-3 : MATHEMATICS**SECTION-I(i) : (Maximum Marks: 24)**

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

1. Which of the following is/are correct

(A) Let $f(x) = \frac{x^3}{3} + \frac{x^2}{2} + ax + b \quad \forall x \in \mathbb{R}$ then $f(x)$ is invertible iff $a \in \left[\frac{1}{4}, \infty\right)$, $b \in \mathbb{R}$

(B) The fundamental period of $f(x) = \cos(\sin x) + \cos(\cos x)$ is π

(C) The number of real solutions of the equation $x^3 + 1 = 2\sqrt[3]{2x-1}$ is 3

(D) The number of real solutions of the equation $x^3 + 1 = 2\sqrt[3]{2x-1}$ is 1

2. $C_1, C_2, C_3, \dots$ is a system of circles with radii $r_1, r_2, r_3, \dots$ such that $r_1 < r_2 < r_3 \dots$. All the circles have a pair of common tangents $x^2 + 4xy + y^2 = 0$ and the angle between them is acute.

If C_1 touches C_2 , C_2 touches C_3 and so on. If $r_1 = 3$ and $x_k = \text{CiS}\left(\frac{\pi}{r_k}\right)$ then $x_1 \cdot x_2 \cdot x_3 \cdot \dots \infty$ is (where $\text{CiS}\theta = \cos\theta + i\sin\theta$)

(A) i

(B) 0

(C) -i

(D) 1

Space for Rough Work

3. If A is an orthogonal matrix ($AA^T = I$) of order n, then $|\text{adj}(\text{adj } A)|$ may be
 (A) 1 (B) -1 (C) 0 (D) $\pm 2^{n^2}$
4. In triangle ABC (with usual notations), if circumradius is 8 times inradius and $B - C = \frac{2\pi}{3}$, then
 (A) $\sin \frac{A}{2} = \frac{1}{4}$ (B) $\frac{b}{c} = \frac{3+\sqrt{5}}{4}$ (C) $\frac{b}{c} = \frac{3+\sqrt{5}}{2}$ (D) $\sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2} = \frac{1}{32}$
5. Which of the following is/are false.
 (A) If $|Z| = \text{Max}\{|Z-1|, |Z+1|\}$ where 'Z' is a complex number then $|Z + \bar{Z}| = 1$
 (B) The number of complex numbers 'Z' such that $Z^2 + Z|Z| + |Z|^2 = 0$ is 2 (when $|Z| \neq 0$)
 (C) If $\left|Z + \frac{2}{Z}\right| = 2$ then maximum value of $|Z|$ is $\sqrt{3} + 2$
 (D) If the roots of $z^3 + iz^2 + 2i = 0$ represent the vertices of ΔABC in argand plane then area of ΔABC is '1' sq. unit
6. If $0 < \alpha, \beta < \pi$, $\lim_{x \rightarrow 0} \left(\frac{(\sin \alpha)^x + (\sin \beta)^x}{2} \right)^{\frac{1}{x}} = 1$, then (where $[.]$ denotes GIF)
 (A) The value of $\tan\left(\frac{\alpha + \beta}{3}\right) = \sqrt{3}$ (B) $[\alpha + \beta] = 3$
 (C) $[\alpha + \beta] = 2$ (D) The value of $\tan\left(\frac{\alpha + \beta}{3}\right) = \frac{1}{\sqrt{3}}$

Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 12)

- This section contains **FOUR** questions.
- **Each question has matching lists.** The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

7. Match the following

List - I

List - II

(P) The number of pairs (x, y) of real numbers satisfying

(1) 9

$$|\tan \pi y| + \sin^2 \pi x = 0 \text{ and } x^2 + y^2 \leq 2 \text{ is}$$

(Q) Let 'M' be the number of solutions of the equation

(2) 7

$$|\sin 2x| = |\sin 3x| \text{ in } [-7\pi, 7\pi] \text{ then } \frac{M-1}{10} =$$

(R) If $[\cos x] + [\sin x + 1] = 0$ then maximum value of $f(x) = \tan x$ is [where $[\cdot]$ denotes G.I.F]

(3) 5

(S) If $[\sin x] + [\sqrt{2} \cos x] = -3$, where $x \in [0, 2\pi]$

(4) 0

Then $[\sin 2x]$ equals [where $[\cdot]$ denotes G.I.F]

(A) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 4$

(B) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 3; S \rightarrow 3$

(C) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 3$

(D) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 4; S \rightarrow 4$

Space for Rough Work

8. Match List-I with List-II

List - I

List - II

(P) A rectangle whose sides are a and b of perimeter 36 cm which sweeps out the largest volume when revolved about one of its sides then $|a - b|$ is equal to

(1) 6

(Q) If $p \in \left(0, \frac{1}{e}\right)$ then the number of distinct roots of the equation $|\ln x| - px = 0$ is

(2) 2

(R) The exhaustive set of values of ' a ' for which the

(3) 1

function $f(x) = \frac{a}{3}x^3 + (a+2)x^2 + (a-1)x + 2$ ($a > 0$) possesses a

negative point of minimum is (q, ∞) the value of q is

(S) The number of integral value of ' a ' for which

(4) 3

$f(x) = x^4 + ax^3 + \frac{3x^2}{2} + 1$ is concave up along

entire real line is

(A) $P \rightarrow 1; Q \rightarrow 4; R \rightarrow 3; S \rightarrow 4$

(B) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 4; S \rightarrow 3$

(C) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 4$

(D) $P \rightarrow 1; Q \rightarrow 4; R \rightarrow 4; S \rightarrow 3$

Space for Rough Work

9. Match the list

List - I

(P) If l, m, n are 3 consecutive positive odd numbers then the family of lines $lx - my + n = 0, (l < m < n)$ pass through then point

(Q) The centre of the circle through $(0, 2)$ and touching $y = x^2$ at $(2, 4)$ is

(R) If $(1, 2), (4, 4)$ are two conormal points from a point

to $y^2 = 4x$, then the third conormal point is

(S) The line $2x + 3y = 12$ is a tangent to the ellipse

$$\frac{x^2}{9} + \frac{y^2}{4} = 2 \text{ at}$$

List - II

(1) $(9, -6)$

(2) $(3, 2)$

(3) $\left(-\frac{2}{3}, \frac{14}{3}\right)$

(4) $(1, 2)$

(A) $P \rightarrow 3; Q \rightarrow 4; R \rightarrow 2; S \rightarrow 1$

(B) $P \rightarrow 4; Q \rightarrow 3; R \rightarrow 1; S \rightarrow 2$

(C) $P \rightarrow 3; Q \rightarrow 4; R \rightarrow 1; S \rightarrow 2$

(D) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 3; S \rightarrow 2$

Space for Rough Work

10. Match the following :

List - I

List - II

(P) An urn contains five balls, two balls are drawn

(1) $\frac{3}{16}$

and are found to be white. The probability of all the balls in urn are white is

(Q) Out of 15 consecutive integers three are selected

(2) $\frac{7}{8}$

at random, then the probability of the sum is divisible by 3 is

(R) If 3 cards are placed at random and independently

(3) $\frac{1}{2}$

in 4 boxes lying in a straight line. Then the probability of the cards going into 3 adjacent boxes such that each box contain one card, is

(S) A box contains 4 balls which are either red or black,

(4) $\frac{31}{91}$

2 balls are drawn and found to be red if these are replaced, then the probability that next draw will result a red ball is

(A) $P \rightarrow 3; Q \rightarrow 4; R \rightarrow 2; S \rightarrow 1$

(B) $P \rightarrow 4; Q \rightarrow 3; R \rightarrow 1; S \rightarrow 2$

(C) $P \rightarrow 3; Q \rightarrow 4; R \rightarrow 1; S \rightarrow 2$

(D) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 3; S \rightarrow 2$

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.
For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

		+			
●	●	○	○	○	○
①	①	①	①	①	①
②	②	②	●	②	②
③	③	③	③	③	③
④	④	④	④	④	④
⑤	⑤	⑤	⑤	⑤	●
⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	●	●	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨

		-			
●	●	○	○	○	○
①	①	①	①	①	①
②	②	②	②	②	②
③	③	③	③	③	③
④	④	④	④	④	④
⑤	⑤	⑤	●	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

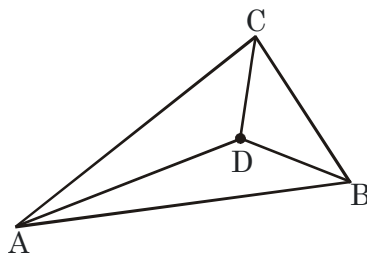
1. If $\int_{-\frac{\pi}{4}}^{\frac{3\pi}{4}} \frac{e^{\frac{\pi}{4}}}{\left(e^x + e^{\frac{\pi}{4}}\right)(\sin x + \cos x)} dx = \lambda \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sec x dx$, then $\frac{1}{16\lambda^2}$ is equal to

2. If $f(x) = \begin{vmatrix} \sin(x + \alpha) & \sin(x + \beta) & \sin(x + \gamma) \\ \cos(x + \alpha) & \cos(x + \beta) & \cos(x + \gamma) \\ \cos(\beta - \gamma) & \cos(\gamma - \alpha) & \cos(\alpha - \beta) \end{vmatrix}$ and $f(9) = \lambda (\lambda \neq 0)$, then find the value of $\frac{\sum_{k=1}^9 f(k)}{\lambda}$.

3. Four letters, two 'a' and two 'b' are filled into 16 cells of a matrix as given. It is required that each cell contains atmost one letter and each row or column cannot contain same letters. Then the number of ways the matrix can be filled is λ then sum of digits of λ is

Space for Rough Work

4. The line $x = 0$ divides the area enclosed by the curves $|x - 1| - y = 0, |x| + y - 3 = 0$ into two areas R_1 and R_2 . Then the value of $R_1 + R_2$ is
5. A differentiable function 'f' is satisfying the relation $f(x + y) = f(x) + f(y) + 2xy(x + y) - \frac{1}{3}, \forall x, y \in \mathbb{R}$ and $\lim_{h \rightarrow 0} \frac{3f(h) - 1}{6h} = \frac{2}{3}$. then find the value of $f(2)$
6. A tetrahedron ABCD shown in the diagram having its faces ABD, ACD, ABC parallel to the planes $x + y = 2, x + z = 2, x + y + z = 4$ respectively



If $AB = \sqrt{2}, AD = \sqrt{3}, AC = \sqrt{2}$ units then the reciprocal of the volume of the tetrahedron in cubic units is

Space for Rough Work

7. If the tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at the point $(a\cos\theta, b\sin\theta)$ meets the auxiliary circle in two points A, B such that the chord AB subtends a right angle at the centre, then the eccentricity of the ellipse is $\frac{1}{\sqrt{\alpha + \beta \sin^2 \theta}}$ where $(\alpha + \beta)^2 =$
8. The value of 'c' for which $|\alpha^2 - \beta^2| = \frac{7}{4}$, where α and β are the roots of $2x^2 + 7x + c = 0$ is
-

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

Space for Rough Work